ASSIGNMENT- 06

Bachelor of Computer Applications (Hons. AI & ML)
Semester- III

Subject: Database Management System

Topic: Implementation of Data Integrity Constraints

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1. Introduction

Integrity constraints are rules applied to a database to maintain the accuracy, validity, and consistency of data. They prevent invalid entries and ensure that relationships between tables remain correct.

In this assignment, integrity constraints are implemented in the **Paranormal Investigation Agency Management System** using SQL in MySQL.

2. Objectives

- To implement primary key and foreign key constraints.
- To apply `UNIQUE`, `CHECK`, and `DEFAULT` constraints.
- To maintain data accuracy and consistency.
- To test the constraints using SQL commands.

3. Types of Integrity Constraints

3.1 Primary Key Constraint

A primary key uniquely identifies each record in a table. It does not allow duplicate or NULL values.

Example:

client_id INT PRIMARY KEY AUTO_INCREMENT

Explanation:
The `client_id` column uniquely identifies each client in the Clients table.

3.2 Foreign Key Constraint

A foreign key establishes a relationship between two tables and ensures that a referenced record exists.

Example:

FOREIGN KEY (Client_id)

REFERENCES Clients(client_id)

Explanation:
This constraint ensures that every client referenced in the Cases table exists in the Clients table.

3.3 UNIQUE Constraint

The `UNIQUE` constraint prevents duplicate values in a column or combination of columns.

Example:

Case_id INT NOT NULL UNIQUE

Explanation:
In the Reports table, this constraint ensures that a case cannot have more than one report.

3.4 CHECK Constraint

The `CHECK` constraint ensures that values satisfy a specified condition.

Example:

CHECK (Urgency_level IN ('Low', 'Medium', 'High'))

Explanation:
This constraint allows only Low, Medium, or High as urgency levels.

3.5 DEFAULT Constraint

The `DEFAULT` constraint automatically assigns a value when no value is provided for a column during insertion.

Example:

Case_status VARCHAR(50) DEFAULT 'Reported'

Explanation:
If no case status is provided, the database automatically assigns the status `Reported`.

3.6 NOT NULL Constraint

The `NOT NULL` constraint ensures that a column cannot contain a NULL value.

Example:

name VARCHAR(100) NOT NULL

Explanation:
Every client record must contain a name.

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4. Implementing Integrity Constraints

The following SQL commands add additional integrity constraints to the tables created in Assignment 5.

4.1 Apply UNIQUE Constraints

```
USE Paranormal_Investigation_Agency;


ALTER TABLE Clients

ADD CONSTRAINT uq_clients_email UNIQUE (email);


ALTER TABLE Investigators

ADD CONSTRAINT uq_investigators_email UNIQUE (email);
```

Explanation:
These constraints prevent the same email address from being entered for multiple clients or investigators. MySQL permits multiple NULL values in a UNIQUE column.

4.2 Apply CHECK Constraints

```
ALTER TABLE Cases

ADD CONSTRAINT chk_case_status

CHECK (Case_status IN

('Reported', 'Under Investigation', 'Closed', 'Unresolved'));


ALTER TABLE Cases

ADD CONSTRAINT chk_urgency_level

CHECK (Urgency_level IN ('Low', 'Medium', 'High'));
```

```
ALTER TABLE Investigators
ADD CONSTRAINT chk_experience_level
CHECK (experience_level IN
('Trainee', 'Field Agent', 'Lead Investigator'));

ALTER TABLE Equipment
ADD CONSTRAINT chk_condition_status
CHECK (condition_status IN
('Working', 'Under Repair', 'Retired'));

ALTER TABLE Reports
ADD CONSTRAINT chk_final_verdict
CHECK (Final_verdict IN
('Confirmed Activity', 'Debunked', 'Inconclusive'));

ALTER TABLE Payments
ADD CONSTRAINT chk_payment_status
CHECK (Payment_status IN ('Pending', 'Paid', 'Refunded'));

ALTER TABLE Payments
ADD CONSTRAINT chk_payment_amount
CHECK (Amount >= 0);
```

Explanation:
These constraints restrict the values entered into the database to the permitted options. The payment amount cannot be negative.

4.3 Apply DEFAULT Constraints

ALTER TABLE Cases

ALTER COLUMN Case_status SET DEFAULT 'Reported';

ALTER TABLE Cases

ALTER COLUMN Urgency_level SET DEFAULT 'Medium';

ALTER TABLE Equipment

ALTER COLUMN condition_status SET DEFAULT 'Working';

ALTER TABLE Payments

ALTER COLUMN Payment_status SET DEFAULT 'Pending';

Explanation:
These commands provide default values when the corresponding columns are omitted from an INSERT statement.

MySQL note: If your MySQL version does not support `ALTER COLUMN ... SET DEFAULT`, use the following form instead:

ALTER TABLE Cases

ALTER Case_status SET DEFAULT 'Reported';

ALTER TABLE Cases

ALTER Urgency_level SET DEFAULT 'Medium';

ALTER TABLE Equipment

ALTER condition_status SET DEFAULT 'Working';

ALTER TABLE Payments

ALTER Payment_status SET DEFAULT 'Pending';

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5. Testing Integrity Constraints

The following commands demonstrate how the database responds to valid and invalid data. Execute the valid tests using existing parent records in your database.

5.1 Test the Primary Key Constraint

INSERT INTO Clients

(client_id, name, email, phone, address, registration_date)

VALUES

(1, 'Alex Morgan', 'alex@example.com', '9876543210',

'Bengaluru', '2026-10-01');

Explanation:
This statement inserts a client with ID 1, provided that the ID does not already exist.

Invalid test: Attempting to insert another client with the same `client_id` should fail because primary key values must be unique.

5.2 Test the Foreign Key Constraint

INSERT INTO Cases

(Client_id, Location_id, Category_id, Date_reported,

Case_status, Urgency_level)

VALUES

(9999, 1, 1, '2026-10-10', 'Reported', 'High');

Expected result: The insertion should fail if client ID 9999 does not exist in the Clients table. It may also fail if location ID 1 or category ID 1 does not exist.

Explanation:
The foreign key constraint prevents a case from referencing a nonexistent client, location, or category.

5.3 Test the UNIQUE Constraint

INSERT INTO Clients

(name, email, phone, address, registration_date)

VALUES

('Jordan Lee', 'alex@example.com', '9876500000',

'Chennai', '2026-10-02');

Expected result: If `alex@example.com` already exists in the Clients table, this insertion should fail because the email address must be unique.

5.4 Test the CHECK Constraint

INSERT INTO Cases

(Client_id, Location_id, Category_id, Date_reported,

Case_status, Urgency_level)

VALUES

(1, 1, 1, '2026-10-10', 'Invalid Status', 'High');

Expected result: The insertion should fail because `Invalid Status` is not an allowed case status. The referenced client, location, and category must also exist.

5.5 Test the DEFAULT Constraint

INSERT INTO Cases

(Client_id, Location_id, Category_id, Date_reported)

VALUES

(1, 1, 1, '2026-10-10');

Expected result: If the referenced records exist, the case should be inserted with the default status `Reported` and urgency level `Medium`.

To verify the result:

SELECT Case_id, Case_status, Urgency_level

FROM Cases

ORDER BY Case_id DESC

LIMIT 1;

5.6 Test the NOT NULL Constraint

INSERT INTO Clients

(name, email, phone, address, registration_date)

VALUES

(NULL, 'test@example.com', '9876501111',

'Bengaluru', '2026-10-10');

Expected result: The insertion should fail because the `name` column does not allow NULL values.

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6. Verifying Constraints

The following command displays the table definition and its constraints.

SHOW CREATE TABLE Cases;

To inspect the constraints defined in the current database:

SELECT

TABLE_NAME,

CONSTRAINT_NAME,

CONSTRAINT_TYPE

```
FROM information_schema.TABLE_CONSTRAINTS

WHERE CONSTRAINT_SCHEMA = 'Paranormal_Investigation_Agency';
```

Explanation:
These commands help verify that the primary keys, foreign keys, and other constraints have been created successfully.

7. Importance of Integrity Constraints

Integrity constraints are important because they:

1. Prevent duplicate records where uniqueness is required.
2. Ensure that foreign keys refer to existing records.
3. Restrict invalid values.
4. Automatically provide default values when appropriate.
5. Prevent missing values in mandatory columns.
6. Maintain data accuracy and consistency across the database.

8. Result

Integrity constraints were defined and tested for the Paranormal Investigation Agency Management System using MySQL. Primary keys, foreign keys, UNIQUE, CHECK, DEFAULT, and NOT NULL constraints help maintain the validity and consistency of the database.

9. Conclusion

This assignment demonstrated how integrity constraints protect the database from invalid, duplicate, or inconsistent data. Applying these constraints improves the reliability of the Paranormal Investigation Agency Management System and ensures that related records remain correctly connected.